Exercise 2: Unsupervised Exploration – SOLUTIONS

Machine Learning with R in the Life Sciences

Setup

```r
library(palmerpenguins)
library(tidyverse)
penguins_clean <- penguins |>
  drop_na() |>
  select(species, bill_length_mm, bill_depth_mm, flipper_length_mm, body_mass_g)
```

2.1

```r
penguins_numeric <- penguins_clean |>
  select(-species)
pca_result <- prcomp(penguins_numeric, center = TRUE, scale. = TRUE)
```

2.2

```r
summary(pca_result)
```

```
## Importance of components:
##                           PC1    PC2     PC3     PC4
## Standard deviation     1.6569 0.8821 0.60716 0.32846
## Proportion of Variance 0.6863 0.1945 0.09216 0.02697
## Cumulative Proportion  0.6863 0.8809 0.97303 1.00000
```

Q: Why is it useful to know how much variance each PC captures?

The variance explained by each PC tells you **how much information** that component carries. If PC1 captures 69% and PC2 captures 19% of the total variance, then a 2D scatter plot of PC1 vs. PC2 represents about 88% of the total variation in the data — meaning you are getting a faithful low-dimensional summary with very little information loss.

This matters for several practical reasons:

- **Deciding how many PCs to keep:** If the first 3 PCs explain 95% of variance, the remaining PCs mostly capture noise and can be discarded. If you need 50 PCs to reach 80%, the data has many independent dimensions and is harder to compress.
- **Interpreting plots:** A PCA plot where PC1 + PC2 explain 90% of variance is trustworthy — the clusters you see are real. A plot where they explain only 15% is misleading — important structure may be hidden in higher PCs.

- **As a preprocessing step:** When using PCA for dimensionality reduction before ML (e.g., `step_pca(num_comp = 10)`), the variance explained helps you choose how many components to retain so you keep enough signal without flooding the model with noise dimensions.

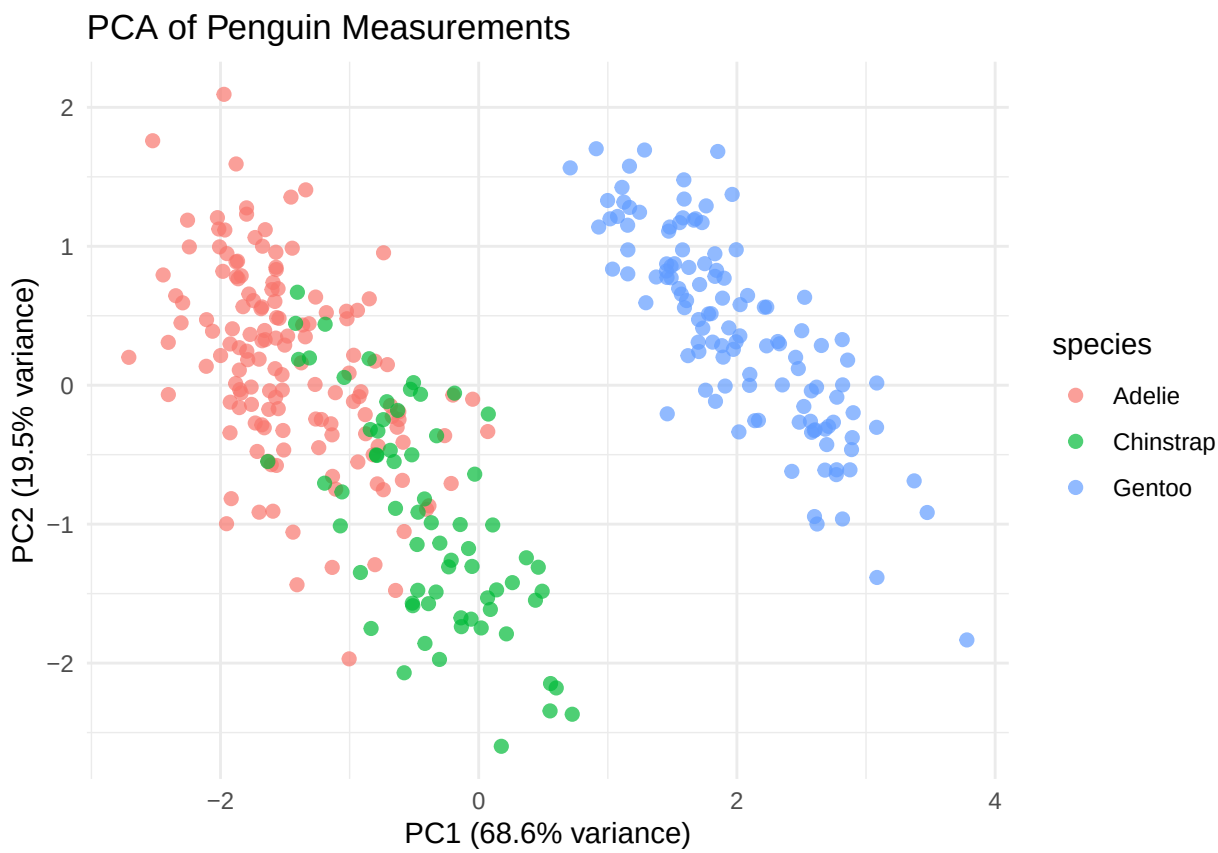
Q: Do the species form distinct groups in PC space? Which species is most separated from the others?

Yes, the species form distinct groups. **Gentoo** is most clearly separated from the others, sitting apart along PC1. This is because Gentoo penguins are substantially larger (higher body mass and flipper length) than both Adelie and Chinstrap, and PC1 primarily captures this size axis — it is the dominant source of variation in the data.

Adelie and Chinstrap overlap more in PC space, particularly along PC1, though they begin to separate along PC2 (which captures more of the bill shape differences). This tells us that a classifier would easily distinguish Gentoo from the other species, but would need to rely on subtler feature differences (bill dimensions) to tell Adelie and Chinstrap apart.

2.3

```r
pca_data <- data.frame(
  PC1 = pca_result$x[, 1],
  PC2 = pca_result$x[, 2],
  species = penguins_clean$species
)
ggplot(pca_data, aes(x = PC1, y = PC2, color = species)) +
  geom_point(alpha = 0.7, size = 2) +
  theme_minimal() +
  labs(title = "PCA of Penguin Measurements",
       x = paste0("PC1 (", round(summary(pca_result)$importance[2, 1] * 100, 1), "% variance)"),
       y = paste0("PC2 (", round(summary(pca_result)$importance[2, 2] * 100, 1), "% variance)"))
```

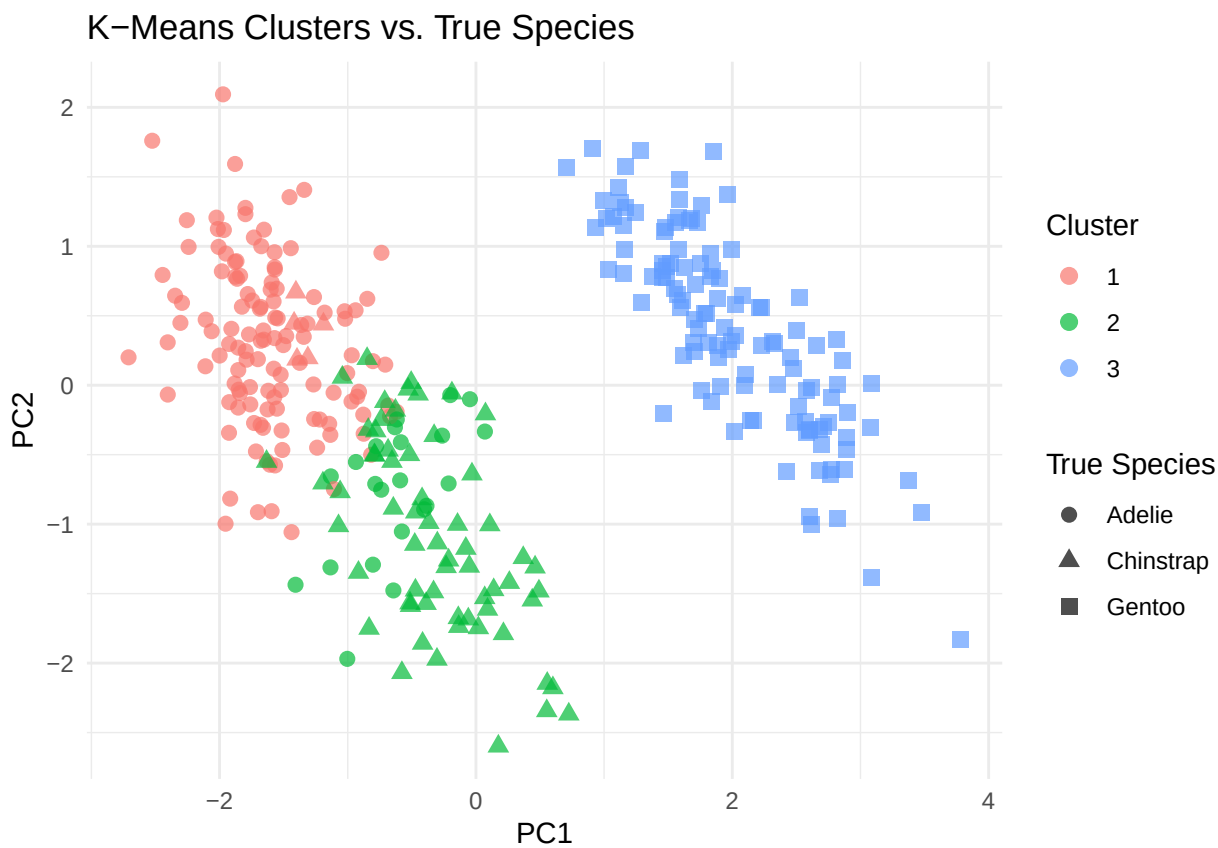


2.4

```
set.seed(42)
kmeans_result <- kmeans(scale(penguins_numeric), centers = 3, nstart = 25)
```

2.5

```
pca_data$cluster <- as.factor(kmeans_result$cluster)
ggplot(pca_data, aes(x = PC1, y = PC2, color = cluster, shape = species)) +
  geom_point(alpha = 0.7, size = 2.5) +
  theme_minimal() +
  labs(title = "K-Means Clusters vs. True Species", color = "Cluster", shape = "True Species")
```



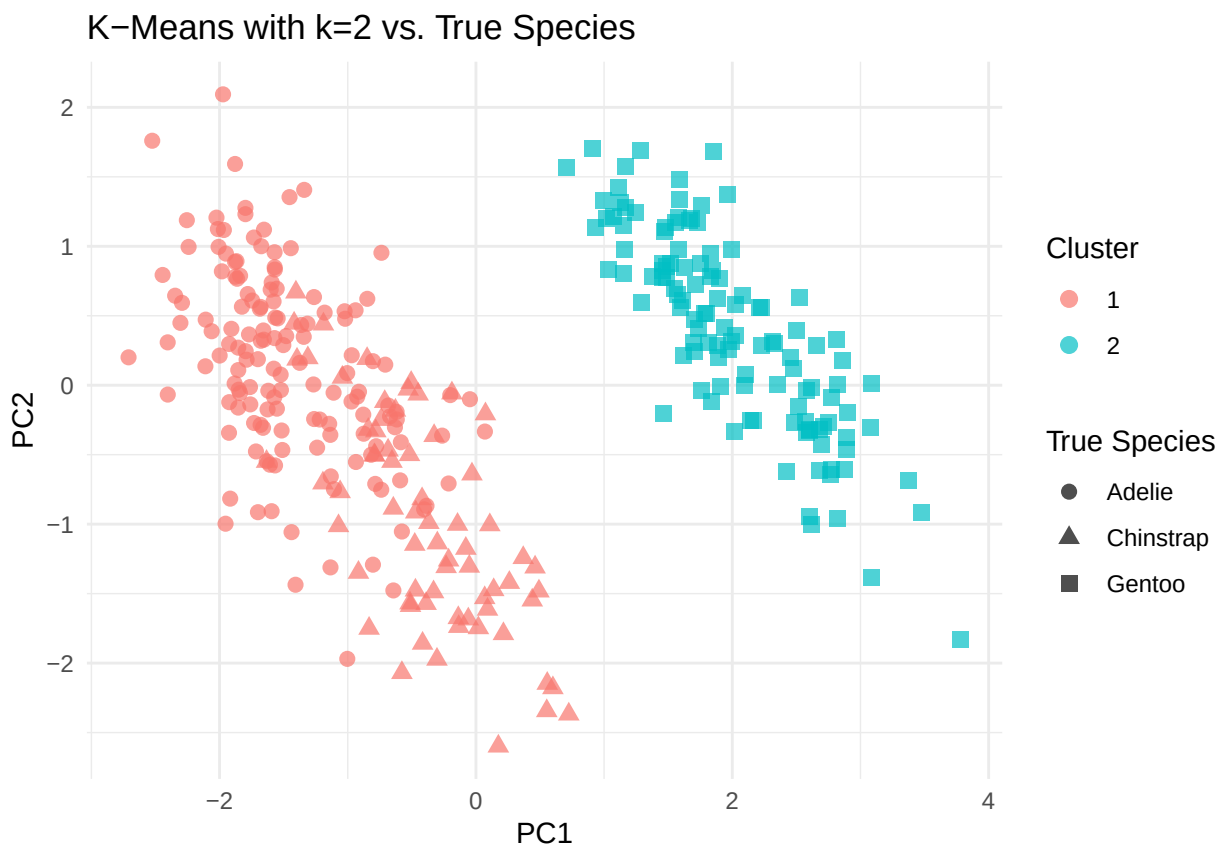
2.6

```
table(True_Species = penguins_clean$species, Cluster = kmeans_result$cluster)
```

```
##              Cluster
## True_Species   1   2   3
##    Adelie    124  22   0
##    Chinstrap   5  63   0
##    Gentoo      0   0 119
```

2.7

```
set.seed(42)
kmeans_k2 <- kmeans(scale(penguins_numeric), centers = 2, nstart = 25)
pca_data$cluster_k2 <- as.factor(kmeans_k2$cluster)
ggplot(pca_data, aes(x = PC1, y = PC2, color = cluster_k2, shape = species)) +
  geom_point(alpha = 0.7, size = 2.5) +
  theme_minimal() +
  labs(title = "K-Means with k=2 vs. True Species", color = "Cluster", shape = "True Species")
```



Discussion Questions — Solutions

Q1: Which two species got merged into one cluster? Does this make sense from the PCA?

With k = 2, **Adelie and Chinstrap** get merged into a single cluster, while Gentoo forms its own cluster. This makes sense from the PCA plot: Gentoo penguins are clearly separated along PC1 (driven mainly by body mass and flipper length), whereas Adelie and Chinstrap overlap considerably in the principal component space.

K-means with k = 2 picks up the most dominant split in the data (Gentoo vs. the rest). This is a useful lesson: unsupervised methods find the strongest structure in the data, which does not always align with the biological labels you care about.

Q2: In a real scenario where you do NOT know the true labels, how would you decide the right value of k?

Without known labels you can use the **silhouette score** (covered in the lecture). For each value of k (e.g., 2 through 10), compute the average silhouette width, which measures how similar each point is to its own cluster compared to the nearest neighboring cluster. The k with the highest average silhouette score is typically a good choice.

Other approaches include:

- **Elbow method:** Plot total within-cluster sum of squares vs. k, look for a "bend."
- **Gap statistic:** Compares within-cluster dispersion to a null reference distribution.

In this penguin dataset, k = 2 will likely have the highest silhouette score because the Gentoo-vs.-rest

split is so clean, even though biologically we know k = 3 is "correct." This highlights an important point: unsupervised methods tell you about data structure, not necessarily biological truth.

Q3: This dataset has only 4 features. In transcriptomics, you might have 20,000 genes. Would PCA be more or less useful in that setting? Why?

PCA becomes **much more useful** in high-dimensional settings. With 20,000 genes you cannot visualize the data directly, and many genes are correlated (e.g., co-regulated in the same pathway). PCA compresses this information into a small number of principal components that capture the dominant variation.

Typically, the first 2–3 PCs already reveal major structure (e.g., tumor vs. normal, batch effects, tissue type). In practice, PCA is one of the first things bioinformaticians run on any new omics dataset — it serves as a quality check:

- Do replicates cluster together?
- Are there outliers?
- Are there batch effects?

Additionally, you can use PCA as a preprocessing step for ML (`step_pca()` in tidymodels), feeding 10–50 PCs into a model instead of 20,000 raw features.